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SYNAPTIC DENSITY IN HUMAN FRONTAL CORTEX — DEVELOPMENTAL CHANGES AND EFFECTS OF AGING

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SUMMARY

Density of synaptic profiles in layer 3 of middle frontal gyrus was quantitated in 21 normal human brains ranging from newborn to age 90 years. Synaptic profiles could be reliably demonstrated by the phosphotungstic acid method (Bloom and Aghajanian[3]) in tissue fixed up to 36 h postmortem. Synaptic density was constant throughout adult life (ages 16–72 years) with a mean of 11.05×10^8 synapses/cu.mm $\pm$ 0.41 S.E.M. There was a slight decline in synaptic density in brains of the aged (ages 74–90 years) with a mean of 9.56×10^8 synapses/cu.mm $\pm$ 0.28 S.E.M. in 4 samples ($P < 0.05$). Synaptic density in neonatal brains was already high — in the range seen in adults. However, synaptic morphology differed; immature profiles had an irregular presynaptic dense band instead of the separate presynaptic projections seen in mature synapses. Synaptic density increased during infancy, reaching a maximum at age 1–2 years which was about 50% above the adult mean. The decline in synaptic density observed between ages 2–16 years was accompanied by a slight decrease in neuronal density. Human cerebral cortex is one of a number of neuronal systems in which loss of neurons and synapses appears to occur as a late developmental event.

INTRODUCTION

Recent studies of fine structural organization of cerebral cortex in brains from the retarded have shown abnormalities of dendritic development in a number of disorders including chromosomal defects[13,14], familial microcephaly[11], cerebral lipidosis[17] and Menke's Kinky Hair Syndrome[16]. Malformations of dendritic trees of cortical pyramidal neurons also appear to be common and severe in infants with the syndrome of infantile spasms and developmental retardation[11]. Modifications of the Golgi method have been used in all of these studies. In some of the abnormal brains, dendritic spines have appeared to be diminished in number, raising the question

whether density of synaptic contacts might be diminished in addition to a decrease in complexity of dendritic arborization. On the other hand, abnormal persistence of basal dendrites on Purkinje cells has been described in Menke's Kinky Hair Syndrome, which might be expected to be associated with a regional increase in synaptic density.

Direct study of synaptic density in human cerebral cortex has been hampered by lack of methods that allow rapid quantitation of synapses, and by lack of normative data, both as regards age-related variations and effects of postmortem autolysis on ability to demonstrate synaptic profiles. The present study was undertaken to assess the usefulness of an electron microscope–histochemical method for demonstration of synapses in human material, and to obtain normal values on synaptic density in human cerebral cortex against which findings in abnormal brains can be compared.

METHODS

Middle frontal gyrus was chosen for study, since considerable developmental data on this region of cortex in the human are available, especially as regards development of dendritic branching in cortical neurons[18]. Cases with known neurologic disease or with severe prolonged hypoxia prior to death were excluded. Small blocks of cortex, including layers 1–3, were removed at autopsy and were fixed in 5% glutaraldehyde in phosphate buffer (0.3 *M*, pH 7.4). The tissue was prepared for electron microscopy by the phosphotungstic acid method of Bloom and Aghajanian[3], which stains synaptic profiles selectively. In brief, the tissue was dehydrated in graded ethanol solutions and was then stained for one hour in 1% phosphotungstic acid in absolute ethanol, containing a trace of water (1–4 drops of 95% ethanol/10 ml of absolute ethanol). The tissue was then equilibrated with propylene oxide and embedded in Spurr low-viscosity media. Thin sections (silver interference color, about 75 nm in thickness) were cut with a Sorvall MT2 microtome, and were examined in an RCA Elmoscope 4 electron microscope. Photographs were taken of layer 3 frontal cortex in a random fashion at magnification of 6800. Finished prints were 2.5 times magnifications of the negatives, for a total magnification of 17,000. Twenty to thirty 8 × 10 inch prints were prepared for each sample, and identifiable synaptic profiles were enumerated in each print. Only profiles with clearly identifiable presynaptic projections, synaptic cleft and dense postsynaptic band were counted. Mean synapse count per cu.mm was calculated for each sample.

The Abercrombie correction[1] was employed to account for the fact that synapses may extend over several thin sections, and a given synapse could therefore be counted in more than one section, i.e.:

$$P = A\left(\frac{M}{L + M}\right)$$

where P = corrected synapse count, A = raw synapse count, M = thickness of section and L = length of synaptic profiles. This correction is valid for spherical objects only, and a further correction factor has to be introduced to account for the

fact that synapses cut at or near cross-section are not counted by the present method. We followed the convention used by Aghajanian and Bloom[2] of assuming that one-half of actually present synapses in a given section lie in planes where they are sufficiently well demonstrated to be counted. Length of synapses was found to be approximately equal in all specimens examined and a mean synaptic length of 241 Å was assumed for the calculations. The total correction factor therefore was:

$$P = A\left(\frac{0.075}{0.075 + 0.241}\right) \times 2 \text{ or } P = 0.47A.$$

Counts of neuronal density were obtained in 7 μm paraffin-embedded sections stained with hematoxylin and eosin or with toluidine blue. The Abercrombie correction was applied as described above.

RESULTS

Samples from 21 brains, at ages ranging from newborn to 90 years, were studied. Synaptic profiles were clearly evident in postmortem cerebral cortical tissue, although their definition was often somewhat less sharp than in perfusion-fixed experimental tissue (Fig. 1A). The presynaptic projections and postsynaptic bands appeared somewhat more diffuse and less sharply demarcated, and the intracleft lines were not always demonstrable. However, these features did not seriously interfere with the ability to enumerate synaptic profiles. Synapse counts appeared to be stable for 35 h postmortem (Fig. 2), no relationship being demonstrable between calculated synaptic densities and the length of time that elapsed before the tissue was fixed in glutaraldehyde.

Synapses were distributed fairly evenly throughout layer 3 of cerebral cortex. In counts of about 20 prints from the same cortical area, the standard deviation varied from 2 to 6 synapses per print, while the mean number of synapses per print varied from 11.9 to 23.1 in samples from different brains.

Synaptic density in middle frontal gyrus appeared to be stable throughout adult life from age 16 to 72 years (Fig. 3). It varied between 8.9 and 12.5 $\times$ 10^8 synapses/cu.mm with a mean of 11.05 $\pm$ 0.41 S.E.M. (n = 8): Synaptic density in a cortical sample from a 54-year-old man fell somewhat below the range of other adult brains. This sample was from a derelict and long-term heroin addict. However, the history did not provide any evidence of impairment in intellectual functioning. A slight decline in synaptic density was observed past age 72 years, most marked in the oldest brain examined, from a 90-year-old man. Mean synaptic density in 4 brains from persons aged 74–90 years was 9.56 $\times$ 10^8 synapses/cu.mm $\pm$ 0.28 S.E.M. The difference between this group and the 16–72 year group is significant at the 5% level ($t = 2.38$, $P < 0.05$). While this raises the question whether cortical synapses may be lost in a diffuse manner with age, any such effect appears to be slight.

Marked age-related differences in synaptic profiles were found during childhood. Synaptic density at birth was already fairly high, in the range found in the adult.



Fig. 1. A: two synaptic profiles in layer 3 of human middle frontal gyrus, adult case. Phosphotungstic acid stain. Original magnification × 15,000. The presynaptic projection, intracleft line and postsynaptic band are well demonstrated. B: synaptic profile from middle frontal gyrus of an 8-month-old infant. The profile has a nearly symmetrical appearance, with a continuous, irregular presynaptic band. Original magnification × 30,000.

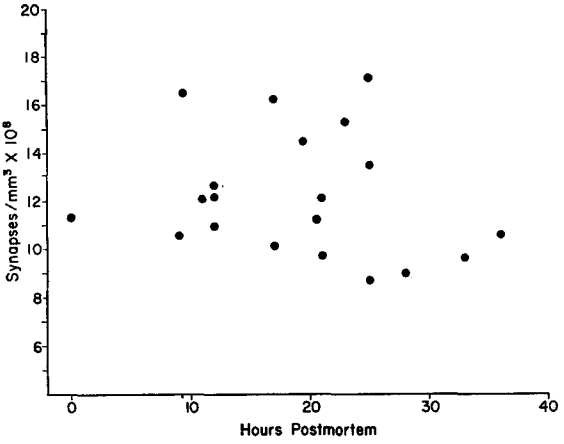


Fig. 2. Synapse counts in layer 3 of middle frontal gyrus plotted against interval between time of death and fixation of the tissue.

However, the immature synaptic profiles were clearly distinguishable from the adult on the basis of morphologic criteria. Separation of the dense presynaptic projections into well defined polygonal shapes was not yet evident. Instead there was a continuous presynaptic band of irregular width, parallel to the postsynaptic band (Fig. 1B). At times, the immature synaptic profiles had a nearly symmetrical appearance. Synapses gradually took on adult morphologic characteristics between ages 6 and 24 months.

Synaptic density in brains of infants and young children between ages 6 months and 7 years was 14.8 $\times$ 10^8 synapses/cu.mm with a range of 13.5 to 16.5 $\times$ 10^8 (n = 7). The difference between synaptic density in the childhood group (ages 6 months to 7 years) and in the adult (16 years to 72 years) was statistically significant ($t = 6.64$, $P < 0.001$).

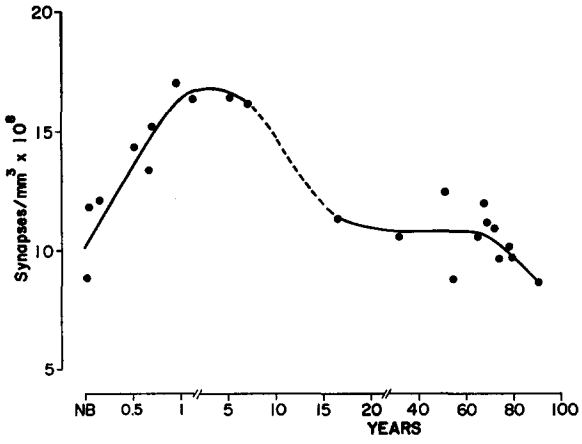


Fig. 3. Synapse counts in layer 3 of middle frontal gyrus as a function of age.

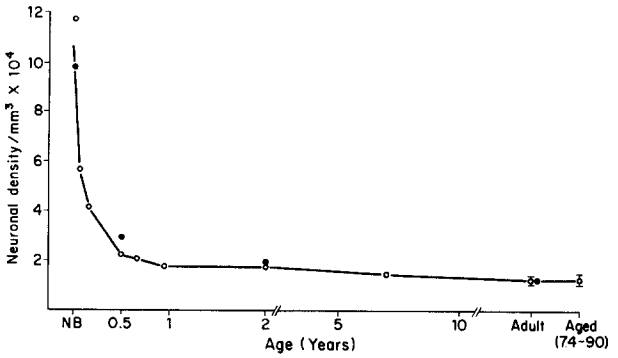


Fig. 4. Neuronal density in layer 3 of middle frontal gyrus as a function of age. Open circles = data derived from the present study. Closed circles = data obtained by Schade and Van Groenigen[18]. Confidence limits = ± 1 S.D.

Neuronal density in layer 3 of middle frontal gyrus also showed marked age-related variations. It was high in neonatal brain (about 10^5 neurons/cu.mm) and fell very rapidly during the first 6 months of life (Fig. 4). Between ages 1 and 2 years neuronal density remained unchanged at about 2×10^4/cu.mm. Subsequently, there was a further, more gradual decrease. Density of neurons in layer 3 at age 2 years was about 55% above the adult mean of $1.32 \times 10^4 \pm 0.04$ S.E.M. In the brain from a 7-year-old, neuronal density was about 10% above the adult mean, but fell within 2 standard deviations of the adult mean.

Two samples from abnormal brains, not included in the above analysis, were also studied. One was the brain of a microcephalic infant with multiple congenital anomalies who died at age 2.5 months of complications of congenital heart disease. This brain weighed 325 g (mean normal for age 489 g) but aside from its small size and somewhat simple gyral pattern, it showed no gross or light microscopic lesions. Synaptic density in middle frontal gyrus was low in this case, with a mean of 6.13×10^8 synapses/cu.mm (± 0.32 S.E.M.). Two normal brains of similar age had mean synapse counts of 11.5 and 12.0×10^8, almost double the number found in the abnormal brain. The second abnormal brain was that of a 6 year old boy who died of tuberous sclerosis complicated by a large periventricular glioma. The child had been in the trainable range of mental retardation and had had numerous grand mal seizures with poor response to anticonvulsant medication. Synaptic density in middle frontal gyrus of this case was unusually high, with a mean synapse count of 18.75 ± 0.67 S.E.M. Details of histologic findings in this case, including results of study by the Golgi method, will be presented elsewhere.

DISCUSSION

The present study indicates that synaptic profiles can be demonstrated in human postmortem brain tissue by use of the phosphotungstic acid method. The method has the advantage over conventional electron microscopy that other structures are virtually unstained. Synapses stand out against a fairly homogeneous background and

can be enumerated with ease in relatively low magnification prints. The synaptic density values obtained with this method in human cerebral cortex are somewhat higher than those reported by Cragg[6], who used conventional electron micrographs for his synapse counts; i.e., a mean synaptic density of about 6×10^8 per cu.mm, reported by Cragg in adult brain, versus a mean of 11×10^8 in the present study. This difference may be related to differences in assumptions regarding correction of absolute counts, to differences related to estimation of thickness of sections and to difficulties encountered in identifying all synapses during screening of osmium tetroxide–uranyl acetate treated brain sections. Furthermore, Cragg's data are based on counts in various cortical areas, while only layer 3 of middle frontal gyrus was examined in the present study. It is likely that differences in synaptic density exist in different cortical layers, as well as in different regions of cortex.

In agreement with findings reported by Cragg[6], no significant decline in synaptic density in frontal cortex was observed in the present study over the adult age range of 16–72 years. Further data are needed to determine whether a slight decrease in synaptic density observed in 4 brains above age 74 years represents a significant finding. Neuronal density in frontal cortex of the 4 aged brains did not differ from that found in younger adults, nor was there a difference in total brain weights. This is at variance with prior reports of reduction in brain weight and of neuronal loss with aging[4]. However, the small reductions that have been reported, usually in the range 5–10% from young adult values, could easily have been missed in this small sample. Moreover, the aged brains that were studied represent a selected group, since cases with known mental or other neurologic changes were excluded. The differentiation of effects of age per se from diseases of the brain associated with aging presents major problems in this age group.

The significantly greater density over adult values that was observed during late infancy and childhood was a somewhat unexpected finding. Studies on synaptic development in animals have shown a similar trend, but of somewhat smaller magnitude. In the rat brain, Aghajanian and Bloom[2] found a synaptic density at age 35 days that was about 10% higher than synaptic density in the adult. Cragg[7], studying visual cortex of the kitten, found that synaptic density at age 36 days was about 50% above adult level. It is of interest that synaptic density in kitten cerebral cortex continued to be above the adult value as late as age 15 weeks, at a time when the kitten approaches maturity. This finding confirms the fact that synaptic density in mammalian cerebral cortex declines late in development, after brain growth is nearly complete.

Comparison of the developmental data in the human with those of Bloom and Aghajanian in the rat underscores the marked differences in cortical development at birth between these two species. Synaptic profiles were not demonstrable at birth in rat cerebral cortex by the PTA method, while they are already very numerous in neonatal human isocortex. The morphology of immature synapses in man resembles that reported in cerebral cortex of the rat. Immature synapses stained with PTA do not have clearly distinct presynaptic projections. Instead, there is a continuous presynaptic band which imparts an almost symmetrical appearance to the profiles. The significance of this finding is presently unknown.

Changes in synaptic density in neocortex during infancy and childhood are best analyzed in the context of what is known about packing density and dendritic growth of neurons in this region of cerebral cortex. Values on packing density derived from the present study resemble those reported by Schade and Van Groenigen in the same cortical area (Fig. 4). The density of neurons declines very markedly in the first 6 months of postnatal life and a slower decline continues between age 2 years and maturity. Using data concerning packing density of neurons together with data on synaptic density, it is possible to calculate an approximate mean number of synapses/neuron in layer 3 of middle frontal gyrus (Fig. 5). This value increases rapidly from about 10,000 at birth to 100,000 at age one year. A slight decrease in the number of synapses per neuron appears to occur between late chilhood and the adult. A similar decrease in number of synapses per neuron was noted by Cragg[7] in visual cortex of the cat between age 15 weeks and adult.

The data suggest that neurons acquire their full complement of synapses by age one year, and that some synaptic loss occurs subsequently. In contrast, dendritic arborization of pyramidal neurons in the same area appears to increase considerably past age one year. Schade and Van Groenigen[18] report an approximate doubling of mean dendritic length of pyramidal cells in layer 3 between ages 2 years and adult. These data are based on study of only two brains. However, if accurate, they imply a gradual decline of synaptic density along the dendritic surface, as dendrites continue to expand, and as synapse number per neuron declines. Such changes in spatial relationships of synapses may well form the anatomical substrate for changes in electrical properties of cerebral cortex that occur in late childhood, including maturation of electroencephalographic patterns and increase in seizure threshold.

On the basis of our present knowledge, it is possible to subdivide postnatal cortical development into two phases: phase 1 extends from birth to age one year. It is characterized by rapid decline in neuronal density, by an increase in synaptic density and in number of synapses per neuron, by dendritic growth, and by expansion in the total volume of cerebral cortex. Phase two extends from age one year to adolescence. It is characterized by a slow decline in both synaptic and neuronal density. Dendritic growth continues, and the density of synapses along dendrites declines. One unresolved question during this second developmental period concerns the fate of the neurons and synapses that are lost. During phase 1, there clearly is expansion of total cortical volume to account for much of the decrease in cellular density. Some increase in volume of cortical gray matter may occur even after age 2 years. However, to accommodate the loss of neurons per unit volume which occurs between ages 2 years and the adult, one would have to postulate a 50–60% increase in the volume of cortical gray matter, assuming that events in middle frontal gyrus reflect those in other parts of neocortex. While no good estimates of volume of gray matter are available, it is unlikely that such a large expansion of cortex occurs late in development. Total brain weight increases only by about 20% between ages 2 years and maturity, and some of this increase is undoubtedly due to an increase in myelin, (Fig. 6). The human brain is practically fully grown at age 7 years, when synaptic density still remained 36% above the adult mean. The question therefore arises whether there is a significant amount of

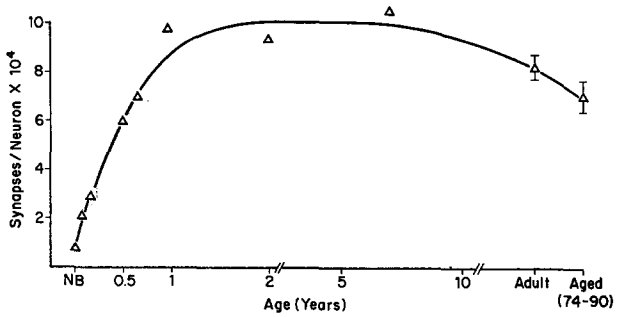


Fig. 5. Mean number of synapses per neuron, derived from data of the present study, as a function of age. Confidence limits = ± 1 S.D.

neuronal and synaptic loss late during cortical development. Neuronal loss as a late developmental event has been described in other systems, such as the anterior horn cells of spinal cord[5,9,10,12,15]. It has been postulated that cells that fail to make functional synaptic contacts undergo degeneration. Such a process, while eliminating unnecessary redundancy, may also remove some of the plasticity that is characteristic of immature brain[10]. Late neuronal and synaptic loss in human cerebral cortex may account for loss of ability to recover from certain types of brain injury, such as from aphasia after dominant hemisphere lesions. Ability to recover speech decreases markedly late in chilhood, at about the time when synaptic and neuronal density decline to adult values.

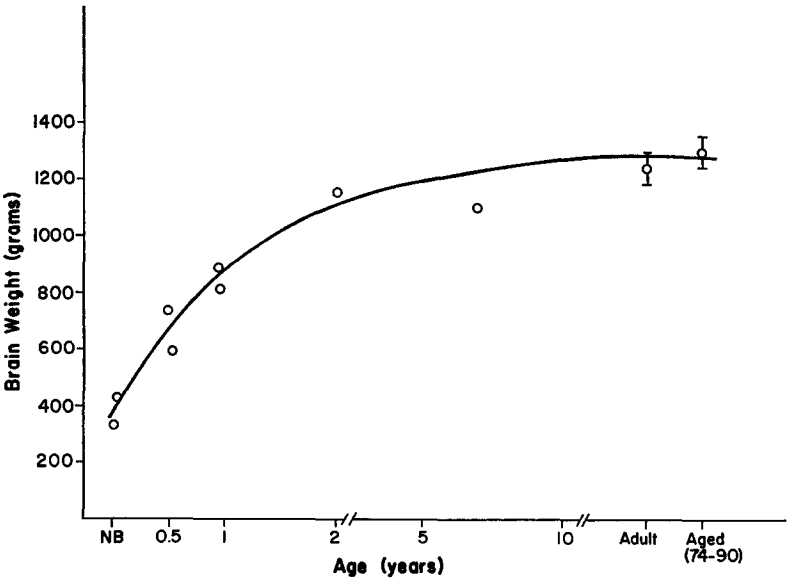


Fig. 6. Total weights of the brains from which the data of the present study were obtained. Confidence limits = ± 1. S.D.

The initial aim of this study was to develop normative data against which values for synaptic density in abnormal brains could be assessed. The data obtained indicate that there is unlikely to be a direct relationship between synaptic density in cerebral cortex and intellect. Certainly, a one-year-old child, whose cerebral cortex may have a synaptic density nearly twice that of a normal adult, has less 'intelligence' as measured by standard tests. A patient whose brain had undergone developmental arrest at age one would be expected to show abnormally high synaptic density as an adult. In this regard, it is of interest that Cragg[6] found statistically significant increases in synaptic density in three brains from retarded individuals. An increase above control values was also noted in the cortex of a child with tuberous sclerosis studied by us. Correlations with age, neuronal density, and dendritic development of neurons are clearly necessary for any meaningful use of synaptic density in the study of abnormal cerebral cortex.

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